MATH 697: Differential Geometry
Section F01 - 2 credits - Spring 2021

Course Information

Instructor Information

Name: David Maxwell
Office: Chapman 308C
Email: damaxwell@alaska.edu
Office Hours: By appointment. To make an appointment, just drop me an e-mail.

Class Information

Dates: January 16, 2021 – May 1, 2021
Location: Web-based via videoconferencing software
Time: Class will meet once per week; meeting time TBA

Prerequisites

MATH 651 (Topology) or permission of the instructor. Recommended maturity in geometry, real analysis, abstract algebra, and familiarity with electromagnetism and quantum mechanics.

Course Description, Goals, and Student Learning Outcomes

This course serves as a rapid introduction to the topology, geometry, and gauge theory underlying the physical theories of electromagnetism and gravity. A successful student will become familiar with the tools they will need in order to properly formulate Maxwell's equations and the Yang-Mills equations on arbitrary spacetimes: manifolds, vector fields, differential forms, cohomology, Lie representations, bundles, connections, curvature, semi-Riemmannian geometry, and more. If time permits, we will explore general relativity with a view towards canonical quantum gravity.

Textbook

Textbook (GFKG): *Gauge Fields, Knots and Gravity*, by John Baez and Javier P Muniain.
https://doi.org/10.1142/2324
Textbook (TGGF): *Topology, Geometry, and Gauge Fields: Foundations*, by Gregory L Nabor.
https://doi.org/10.1007/978-1-4419-7254-5

Course Calendar

Due to the nature of the course, there is no set topics schedule; we will start by studying Part I of GKFG and move to additional chapters at my discretion. Class will meet two hours weekly between Tuesday, January 16 and Monday, April 26 at a time mutually agreed upon by the participants. Below is a *tentative* schedule for the course.

Week	Chapter	Week	Chapter
01/11	TGGF 0—3	03/07	Spring Break
01/17	GFKG I.1-6	03/14	GFKG II.2
01/24	GFKG I.3	03/28	GFKG II.3
01/31	GFKG I.4	04/04	GFKG II.4
02/07	GFKG I.5	04/11	GFKG III.1
02/14	TGGF 3	04/18	GFKG III.2
02/21	GFKG I.6	04/25	Final
02/28	GFKG II.1		

Instructional Methods

Reading: This class will consist mostly of weekly readings from the textbooks.
Lecture: The class will meet weekly to discuss the readings.
Homework: Text exercises may be assigned as homework at my discretion.

Course Policies

e-mail: Check your alaska.edu e-mail account regularly, especially before class meetings. This is the e-mail address to which I have access, and is what I will use to contact you.

Absences: You are expected to attend every meeting. Email me in the event you cannot make a meeting, and be sure to suggest possible times for rescheduling the meeting.

Announcements: All correspondence for the course will occur via email.

Evaluation

This course is graded on a pass/fail basis. The final determination is at my discretion and will be based on your participation in the readings, homework, and class meetings.

Software

You will need a strong command of typesetting software, such as LaTeX, and must be able to communicate mathematics in a live virtual environment, e.g. through the use of a tablet and peripheral devices.

Support Services

You may attend office hours or e-mail me with any questions.

Disabilities Services

The Office of Disability Services implements the Americans with Disabilities Act (ADA) and insures that UAF students have equal access to the campus and course materials. I will work with the Office of Disability Services (203 WHIT, 474-7043) to provide reasonable accommodation to students with disabilities. Please come talk to me as soon as possible if you have/need accommodations.

Student Protections Statement

UAF embraces and grows a culture of respect, diversity, inclusion, and caring. Students at this university are protected against sexual harassment and discrimination (Title IX). Faculty members are designated as responsible employees which means they are required to report sexual misconduct. Graduate teaching assistants do not share the same reporting obligations. For more information on your rights as a student and the resources available to you to resolve problems, please go to the following site: https://catalog.uaf.edu/academicsregulations/students-rights-responsibilities/

COVID-19 Statement

Students are expected to adhere to the university's policies, practices, and mandates and are subject to disciplinary actions if they do not comply. Further, students should keep up-to-date on the university's policies, practices, and mandates related to COVID-19 by regularly checking this website: https://sites.google.com/alaska.edu/coronavirus/uaf/uafstudents?authuser=0